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SUPERSEDING
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MILITARY SPECIFICATION

SHOES, MEN'S, DRESS, OXFORD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a leather blucher oxford shoe.

1.2 Classification. The shoe shall be of one type in the following sizes and widths, as specified in table I below (see 6.2).

TABLE I. Schedule of sizes

Schedule of sizes	4	4½	5	5½	6	6½	7	7½	8	8½	9	9½	10	10½	11	11½	12	12½	13	13½	14	14½	15
XN	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
N	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
R	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8430

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-L-61	- Laces, Nylon
V-T-285	- Thread, Polyester
V-T-295	- Thread, Nylon
KK-I-570	- Insole, Footwear, Leather, Cattlehide
KK-L-165	- Leather, Cattlehide, Vegetable Tanned and Chrome Retanned, Impregnated: And Soles
KK-W-231	- Welting, Leather, Shoe
TT-C-490	- Cleaning Methods For Ferrous Surfaces and Pretreatments for Organic Coatings
ZZ-H-141	- Heels, Rubber
CCC-C-443	- Cloth, Duck, Cotton (Single and Plied Filling Yarns, Flat)
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-C-795	- Cushioning Material, Packaging (Flexible Cellular, Plastic Film) for Packaging Applications

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MIL-L-3931	- Lasts, Footwear, Shoe, Men's, U.S. MIL-1
MIL-C-13924	- Coating, Oxide, Black for Ferrous Metals
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-S-40043	- Soles, Rubber, Shoe
MIL-C-41814	- Counter, Footwear
MIL-L-43283	- Leather, Calfskin, Kip, and Cattlehide for Footwear Uppers, Chrome Tanned
MIL-C-43956	- Cloth, Twill, Cotton, 10.0 ounces (339 g)

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-751	- Stitches, Seams and Stitchings

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 5 - Penetration of Bituminous Materials
D 792 - Specific Gravity and Density of Plastics by Displacement
D 3951 - Standard Practice for Commercial Packaging
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
E 28 - Softening Point by Ring-and-Ball Apparatus

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this document may appear in the sample, in which case this document shall govern.

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3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.3 Materials. (See 6.7).

3.3.1 Leather. All leather components covered by this document shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide.

3.3.1.1 Upper leather. The uppers shall be cut from calfskin, kip, or cattlehide conforming to types I, II, or III of MIL-L-43283. The color of the leather shall be black, including top, base, intermediate, and initial coatings. The top finish applied by the shoe manufacturer shall be a water base, acrylic resin finish or a water base, wax acrylic resin finish (see 4.4.1.1 and 6.5). All finishes applied by the shoe manufacturer and tanner shall be compatible with one another to ensure a long-lasting high luster finish. The finish should be capable of being polished to a high luster with commercial type paste wax before and after normal use. In addition to the defects listed below, any other upper leather defects such as fat wrinkle, or pit and insect damage, plainly visible at 3 feet or more shall be considered as affecting appearance and shall not appear in the vamp or quarters in front of the breast of the heel. Cut parts with scratches, scars, slaughter cuts, damage on the grain surface of the leather, brands, or flesh cuts that show through on the grain surface shall not be used. Flanky, pipey, loose, cracked, or any rough or coarse grain leather shall not appear in the vamps or quarters; however, those defects (with the exception of cracked grain) may appear in the counter area back of the breast of the heel or in the tongue. Clusters of veins, split veins, or pronounced veins shall not appear in the vamps. Vamps shall be given a 100 percent inspection as specified in 4.4.2.1 to ensure elimination of any vamps containing defects listed in 4.4.2.1.

3.3.1.2 Quarter linings. The quarter linings shall be cut from either chrome-tanned or a combination of chrome and vegetable-tanned, full grain or corrected grain calf, kip or side leather, or full grain bellies and shall be not less than 2-1/2 ounces or more than 3-1/2 ounces in thickness when tested as specified in 4.4.1. The leather shall be a natural or light russet vegetable extract color with a smooth finish. A bright pigment finish may be used. As an alternative to the above, the linings may be cut from 2 to 3 ounces thick, full grain chrome tanned or chrome vegetable retanned pigskin. The pigskin shall be drum dyed, and the color shall be light russet. The pigskin shall be finished on the grain side with a light application of aniline finish or on the flesh side with a clear lacquer sealer. The aniline finish or clear lacquer sealer shall be only sufficient to coat the surface fibers of the pigskin and shall not decrease porosity. When the flesh side is finished, the flesh side shall be smooth plated. The finished surface shall be worn next to the foot. The lining leather shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.4.1. Slaughter cuts, flanky, pipey, or off-stretch cuts which may affect serviceability will not be permitted.

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3.3.1.3 Heel pads. The heel pads shall be cut from full grain or corrected grain lining leather, sheepskin, kip or calf leather. The heel pads may also be cut from cattlehide leather splits. The color of the heel pads shall approximate the color of the lining leather. When cattlehide split leather is used, the color shall be imparted by application of a pigmented finish. The leather for the heel pads shall be chrome tanned or combination chrome and vegetable tanned. The thickness of the heel pads shall be 2 to 3-1/2 ounces when tested as specified in 4.4.1. The heel pads shall contain not less than 0.18 percent or not more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.4.1. As an alternative to the above, the heel pads may be cut from the pigskin lining leather specified in 3.3.1.2.

3.3.1.4 Sole leather.

3.3.1.4.1 Outsoles. The outsoles shall conform to the requirements for types III or IV, class 3, style 1, subclass (a) of KK-L-165 and shall be full length with a thickness of 10-1/2 (+1) irons after being fleshed (evened) and impregnated. As an alternative, a black 10 (± 1/2) irons thick rubber outsole conforming to type I, class 1 of MIL-S-40043 may be used. The tread area of the rubber outsole shall be smooth. The outsole may be cut from sheet stock and the size, manufacturer's name or trademark need not be marked on the bottom of the outsoles.

3.3.1.4.2 Insoles. The insoles shall be cut from leather conforming to type I or III, class 1, tannage a or b of KK-I-570. The insoles shall have either a channeled or stuck-on rib (see 3.3.4.6). The thickness of the channeled insoles shall be not less than 5-1/2 irons and not more than 7 irons, and the thickness of the stuck-on rib insoles shall be not less than 4-1/2 irons and not more than 6 irons. As an alternative, the insole may be cut from Texorist material. The Texorist material shall contain 0.25 to 1.00 percent copper-8-quinolinolate uniformly distributed throughout the insole. The Texorist insole shall be 0.116 inch ± 0.010 inch thick. The insole shall be cut with the heel-to-toe direction across the machine direction of the Texorist. The Texorist insole shall have a stuck-on rib applied to the printed side of the material (see 3.3.4.6).

3.3.1.5 Welting. The welting shall conform to the requirements of KK-W-231.

3.3.2 Counters. The counters shall conform to the requirements of MIL-C-41814 for the "Shoes, Men's, Dress, Oxford."

3.3.2.1 Counter sizes. Counters shall be made in eight sizes to cover the range of shoe sizes from 4 to 15 and widths XN to XW. The sizes and shapes shall be defined by patterns furnished by the Government. The counters shall be fitted in accordance with table II.

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TABLE II. Counter fitting schedule

Counter size	1	2	3	4	5	6	7	8
Width:								
XN	5-5½	6-7½	8-9½	10-11½	12-13½	14-15	-	-
N	4-4½	5-6½	7-8½	9-10½	11-12½	13-14½	15	-
R	-	4-5½	6-7½	8-9½	10-11½	12-13½	14-15	-
W	-	4-4½	5-6½	7-8½	9-10½	11-12½	13-14½	15
XW	-	-	4-5½	6-7½	8-9½	10-11½	12-13½	14-15

3.3.3 Adhesives. The adhesives used for binding the various parts of the shoe shall be of the following types.

a. Natural rubber latex
b. Synthetic rubber latex, including chloroprene
c. Natural rubber solvent cement
d. Synthetic rubber solvent cement
e. Synthetic resin cement

3.3.4 Fabrics.

3.3.4.1 Channeled insole reinforcement. The insole reinforcement shall be cotton duck cloth conforming to type I of CCC-C-443.

3.3.4.2 Vamp lining cloth. The vamp lining shall be cut from cotton twill conforming to class 2 of MIL-C-43956.

3.3.4.3 Vamp and quarter doublers (for type I calfskin leather). When type I calfskin leather is used, the vamp and quarter doublers shall be cut from single napped cotton flannel weighing not less than 5.7 ounces per square yard. The finished fabric shall have a texture of 36 ends per inch in the warp and 36 picks per inch in the filling. The breaking strength shall be a minimum of 30 pounds in the warp and 30 pounds in the filling. The starch and protein content including chloroform-soluble and water-soluble material shall be not greater than 15.0 percent based on the weight of the fabric when tested as specified in 4.4.1. The color of the fabric shall be natural (unbleached) or bleached. NOTE: Vamp and quarter doublers are not required when type II kip leather is used.

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3.3.4.4 Quarter top tape. The tape shall be braided using cotton yarns over cotton or nylon warp yarns. The tape shall be 3/32 inch ± 1/32 inch wide and have a minimum breaking strength of 24 pounds when tested as specified in 4.4.1. As an alternative, a bonded monocord nylon thread conforming to type III, size 3 of V-T-295 may be used.

3.3.4.5 Tongue linings. The tongue linings shall be white wool felt, 6 to 7 ounces per square yard, containing a minimum of 35 percent wool when tested as specified in 4.4.1. The water-soluble content shall be a maximum of 4 percent when tested as specified in 4.4.1. As an alternative, the tongue lining may be cut from materials specified in 3.3.4.2 or 3.3.4.3. To minimize ravelling, the tongue lining shall be either cut on the bias direction of the material or be pinked across the top.

* 3.3.4.6 Stuck-on rib (when used, see 3.3.1.4.2). The finished rib shall be 15/64 inch ± 1/64 inch high and a minimum of 1/2 inch in width measured from the inside vertical portion of the rib, and shall extend from heel breast line to heel breast line on the insole. The rib shall be made from coated fabric or a combination of coated fabric and fiberboard. The fabric shall conform to requirements specified in 3.3.4.6.1. The upstanding portion of the rib shall be one of the following: not less than three plies of coated fabric folded over on itself; two layers of coated fabric with a fiberboard filler not less than 0.050 inch thick; or fiberboard and coated fabric.

3.3.4.6.1 Rib construction. The ribbing fabric used in the construction of the rib shall be unbleached cotton or cotton-synthetic fabric conforming to the requirements of table III. The fabric shall be coated on one side with a rubber-base adhesive specified in 3.3.3, and bonded to the flesh side of the insole. The fabric or fiberboard shall cover 5/32 inch (- 2/32, + 1/32 inch) of the peripheral edge and extend over the top of the vertical rib, or the fabric may extend from the inside top of the vertical rib and cover the area between the ribs for a distance not less than 5/8 inch from the inside rib edge. The preformed rib shall be firmly and smoothly fitted and bonded to the flesh side of the insole as specified above to provide the required extension on the finished shoe. As an alternative, the fabric, bonded to the flesh side of the insole between the ribs, is permitted to have a pinked edge.

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TABLE III. Ribbing fabric requirements

Characteristic	Requirement, minimum
Weight	8.0 ounces per sq yd
Yarns per unit length:	
Warp	52 yarns per inch
Filling	30 yarns per inch
Breaking strength:	
Warp	105 pounds
Filling	105 pounds

3.3.4.6.2 Rib strength. The physical requirements for the rib strength shall conform to table IV when tested as specified in 4.4.2.4.

TABLE IV. Rib strength requirements

Characteristic	Minimum 1/	Average
Shear strength	70 pounds	75 pounds
Stitch strength	20 pounds	30 pounds

1/ No single determination (sample unit) shall fall below the minimum value specified, and the average of all specimens (lot average) shall be not less than the average specified.

3.3.4.7 Eyelet stays (for type II kip leather). When type II kip leather is used for the uppers, the eyelet stays shall be a compressible adhesive fiber tape not less than 1/2 inch wide, or a stay of the same minimum width may be cut from 8 ounce stiff starched stay-duck. NOTE: Eyelet stays are not required when type I calfskin leather is used.

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3.3.5 Thread.

3.3.5.1 Thread, upper fitting. The upper fitting thread shall be nylon thread conforming to types I, II, or III, sizes AA, B, and E of V-T-295 except that the requirements for colorfastness shall not apply. The needle thread shall be black and the bobbin or looper thread shall be natural.

3.3.5.2 Thread, sole stitching. The sole stitching thread shall be polyester thread conforming to type I, class 1, subclass C of V-T-285. The machine or running thread shall be black and shall be size 8, 3 ply. The bobbin or shuttle thread shall be natural and shall be size 8, either 3 ply or 5 ply. The requirements for colorfastness shall not apply.

3.3.5.3 Thread, inseaming. The inseaming thread shall be a natural color polyester thread conforming to type I, class 1, subclass C, size 8 (3 or 5 ply) of V-T-285.

3.3.6 Box toes. The box toes shall be cut from a material consisting of a woven base flannel or processed (needled) fabric impregnated with pyroxylin or polystyrene compound. The box toes shall be either an activated or prepared type. The material shall be 0.0875 ± 0.0075 inch thick. The box toes shall conform in shape and size to patterns furnished by the Government and shall be used in the sizes and widths specified in table V. Box toes shall be skived 9/16 inch ± 1/16 inch with a feather edge scarf at breast line only. Lasting allowance of vamp (at toe) must be matched by an identical change in lasting allowance of box toes to provide properly fitting box toes when box toes and vamps are registered at lasting edge. As an alternative, when automated lasting is used, the laminated box toe may be made of virgin ionomer resin (surlyn) laminated with flannel fabric, non-woven fabric, or a combination of both on either or both sides. The total thickness of the alternative material shall be 0.040 inch ± 0.005 inch.

TABLE V. Box toe fitting schedule

Box toe size	1	2	3	4	5	6
Width:						
XN, N	4-6	-	6½-8	-	8½-10	-
R, W, XW	-	4-5½	-	6-7½	-	8-9

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TABLE V. Box toe fitting schedule (cont'd)

Box toe size	7	8	9	10	11
Width:					
XN, N	$10\frac{1}{2}$-$12\frac{1}{2}$	-	13-15	-	-
R, W, XW	-	$9\frac{1}{2}$-11	-	$11\frac{1}{2}$-13	$13\frac{1}{2}$-15

3.3.7 Bottom filler. The bottom filler shall be either thermoplastic or cold process type.

3.3.7.1 Thermoplastic bottom filler. The thermoplastic type shall consist of a mixture of ground cork and a suitable thermoplastic binder in the proportion of a minimum of 2-3/4 parts by volume of cork to each one part of binder. The cork granules shall be free from bark. The ground cork and the binder shall be thoroughly and evenly mixed. The binder shall be water insoluble. The binder shall have a softening point of at least 125°F and a maximum penetration of 85mm with a 200-g load for 60 seconds at 77°F when tested as specified in 4.4.1.

3.3.7.2 Cold process bottom filler. The cold process type shall be spreadable without the use of heat. It shall consist of a mixture of ground cork and a suitable binder in the proportion of two parts cork to one part binder by volume. The cork granules shall be free from bark. When spread filler is dry and set, it shall consist of four parts cork to one part binder. The binder shall be water insoluble and have a softening point of at least 150°F when tested as specified in 4.4.1. The binder shall be tested after evaporation to a constant weight level. As an alternative, a cold process bottom filler applied by semiautomatic equipment or method may be used. The alternative filler shall consist by weight of a mixture of one part ground cork to five parts suitable binder. Upon loss of solvent, the alternative filler shall consist by weight of one part cork to 3-3/4 parts binder. The binder shall be water insoluble and have a softening point of at least 125°F when tested as specified in 4.4.1.

3.3.8 Shank. The shank shall be either a combination shank board and steel shank as specified in 3.3.8.1 or a fiberglass shank as specified in 3.3.8.2.

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3.3.8.1 Steel shank combination. The shank combination shall be made of a shank board piece with a single reverse rib steel shank securely attached by four prongs or two rivets. The steel shank shall be 1/2 inch ± 1/16 inch from the front edge of the shank board piece. There shall be seven sizes of shanks for the run of shoe sizes specified in 3.3.8.3.

3.3.8.1.1 Shank board. The shank board shall be water resistant (shank board that has been hot waxed is considered as being water resistant). The finished thickness of the shank board shall be 1/8 inch ± 1/32 inch. The shank board shall be cut in accordance with Government furnished patterns specified in 3.5. The shank board pieces shall be molded to follow the bottom cavity of the shoes, and both ends shall be skived with a graduated scarf 1-3/8 inches ± 1/16 inch wide. The shank board pieces shall be marked consecutively with a steel stamp starting with figure "1" on the smallest size up to and including "7" on the largest size.

* 3.3.8.1.2 Steel shank. The steel shank shall be constructed from 19 gage 0.0418 inch ± 0.0030 inch cold rolled carbon steel with a hardness ranging from 47 to 54 Rockwell C scale when tested as specified in 4.4.1 and shaped to the manufacturer's standard number 4 bend, conforming to the arch of the MIL-1 last. The width of the steel shank shall be 5/8 inch ± 1/64 inch. The steel shank shall be made with a single rib. The overall thickness of the steel shank with rib shall be 0.080 inch to 0.125 inch. The steel shank shall have a zinc electroplated coating, dull or bright finish, or a zinc compound coating completely and uniformly applied to the base metal following a cleansing by any method of TT-C-490. Steel shanks with black oxide coating conforming to class 1 of MIL-C-13924 or steel shanks with zinc phosphate coating may be used in lieu of steel shanks with zinc coating. The rib shall taper off at a point within 1/2 inch ± 1/8 inch from each end. The steel shank for size 4 shall be 4-1/2 inches ± 1/16 inch in length and the lengths for the other sizes shall grade up or down 1/4 inch ± 1/16 inch.

* 3.3.8.2 Fiberglass shank. The fiberglass shank (see 6.6) shall consist of glass fibers impregnated with vinyl ester resins encased in a plastic sleeve or tube. The width of the uncured encased fiberglass resins shall be 5/8 inch ± 1/32 inch. The shank shall be cured and bonded to the insole and conform to the shape of the last. When fully cured, the shank (glass and resin area) shall be 5/8 inch ± 1/16 inch in width. The forward end of the shank shall have a 1/2 inch ± 1/8 inch taper. The heel end shall be tapered for a minimum of 3/8 inch. The cured shank for size 4 shall be 5 inches ± 1/16 inch in length and the lengths for the other sizes shall be grade up or down 1/4 inch ± 1/16 inch.

3.3.8.3 Shank fitting sizes. The shank fitting shall conform to the schedule as specified in table VI.

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TABLE VI. Shank fitting schedule

Shank size	1	2	3	4	5	6	7
Width:							
XN - N	4-6½	7-8½	9-10½	11-12½	13-14	14½-15	-
R - W	4-5	5½-7	7½-9	9½-11	11½-13	13½-14½	15
XW	4-4½	5-6½	7-8½	9-10½	11-12½	13-14	14½-15

3.3.9 Heels, rubber, washer type. The washer type heel shall conform to type I, class 3, style 1, subclass (a), color No. 1 of ZZ-H-141. The heels shall be used on the shoe sizes shown in table VII.

TABLE VII. Heel schedule

Heel sizes 1/	24	26	28	30	32	34	36	38	40	42
Width:										
XN	5-6	6½-7½	8-9	9½-10½	11-12	12½-13½	14-15	-	-	-
N	4-5	5½-6½	7-8	8½-9½	10-11	11½-12½	13-14	14½-15	-	-
R	4	4½-5½	6-7	7½-8½	9-10	10½-11½	12-13	13½-14½	15	-
W	-	4-4½	5-6	6½-7½	8-9	9½-10½	11-12	12½-13½	14-15	-
XW	-	-	4-5	5½-6½	7-8	8½-9½	10-11	11½-12½	13-14	14½-15

1/ A tolerance of plus one heel size is permitted.

3.3.10 Heel bases. The heel bases shall be rubber or plastic.

3.3.10.1 Rubber heel bases. The molded rubber heel bases shall be black and shall be 3/8 inch ± 1/16 inch high. The hardness of the heel base shall be not less than 83 nor more than 90 when tested as specified in 4.4.1. The heel bases shall not crack when tested as specified in 4.4.1.

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3.3.10.2 Plastic heel bases. The plastic heel base shall be either full (i.e., no center cut out) or a type with a center cut out of normal size. When the center cut out type is used, there shall be a bar positioned across the center of the cut out section. The heel bases shall be designed to ensure they will not crack or fracture during the nailing operation. The use of skelton design (extremely large cut out area) will not be permitted. The heel bases shall be molded from polyethylene resin and the color shall be black. The material shall have a specific gravity between 0.910 and 0.940 and shall be unfilled except for pigment used in coloring. The hardness of the heel bases shall be not less than 44 nor more than 60 when tested as specified in 4.4.1. The heel base shall be free from porosity, cracks, blisters, and other defects affecting serviceability or appearance. The heel base shall be uniform height, 3/8 inch ± 1/32 inch, at all outside edges from back to breast line.

3.3.11 Heel nails. The heel nails shall be steel, cut or wire type, and long enough to produce a clinch on the insole of the shoe. The cut type nails shall be commercial number 450 or 1336 with a 20 gage point. Gaging points shall be done with an English standard wire gage, in which 20 gage is 0.035 inch ± 0.005 inch. The points of the nails should enter the gage at least 1/32 inch. The wire nails shall be commercial type 13-1/2 gage with a flat clinching point.

3.3.12 Tacks. The tacks shall be steel and shall be long enough to thoroughly attach the parts through which they are driven and leave the insoles smooth on the inside.

3.3.13 Pegs and nails, heel seat fastening. The heel seat fastening pegs shall be glue sized paper base fiber, commercial type B or equal, 0.081 inch to 0.111 inch diameter regardless of the type equipment used. The pegs shall be of sufficient length to secure firmly the parts through which they are driven and leave the insole smooth on the inside. Alternatively, the heel seat fastening may be done by using numbers 12 through 39 head, steel or brass nails of sufficient length to clinch on the insole.

3.3.14 Eyelets. The eyelets shall be invisible, plain type, made of aluminum 0.014 inch ± 0.0015 inch thick and shall conform to the finished requirements of table VIII when tested as specified in 4.4.1. After fabrication, the eyelets shall be anodized.

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TABLE VIII. Requirements for finished eyelets

	Mimimum	Maximum
	Inch	Inch
Outside diameter of flange	0.290	0.305
Overall length before setting	0.135	0.150
Diameter of hole before setting	0.120	0.126

3.3.15 Laces, nylon. The laces shall conform to type II, class 1 or 4 of V-L-61. The color of the laces shall be black. The length of the laces shall be 26 inches for shoe sizes 4 to 12, and 27-1/2 inches for shoe sizes 12-1/2 to 15.

3.3.16 Wax, inseaming and sole stitching. The wax shall be white or golden and shall be a permanently plasticized resin. The wax, when in the temperature range of normal machine use, shall thoroughly wax the threads described in 3.3.5.2 and 3.3.5.3.

3.3.17 Ink, edge setting. The ink shall produce a bright hard finish and after setting with a hot iron shall be uniform in appearance. The color of the ink shall be black. When rubber outsoles are used, the ink used for edge finishing shall be black and produce a hard, bright finish. The edges may be padded, brushed or set with a heated iron to obtain a uniform appearance.

3.3.18 Repairers. Repairers shall be liquid spray, or crayon or paste type applied by hand. The color shall match the upper leather and shall have sufficient coverage to correct minor surface imperfections of the leather.

3.3.19 Fillers. Where fillers are used, they shall be capable of providing a foundation for the application of the top finish or edge staining, as applicable, and shall be applied by sponge or spray method.

3.3.20 Top finish. The top finish specified in 3.3.1.1 shall be applied by either the sponge or spray method. The finish shall be a material compatible with and capable of wetting into and binding the previously applied compounds. The finish shall match the dyed leather and shall provide luster.

3.4 Design. The shoe shall be a plain toe, blucher oxford of welt construction as shown on figure 1. The shoe shall have a box toe, cloth vamp lining, leather quarter lining, impregnated leather outsole or synthetic rubber outsole, half rubber heel and molded rubber or plastic heel base.

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3.5 Patterns and dies. Standard sets of component patterns and markers will be furnished by the Government as a basis from which the contractor's dies and patterns shall be made. A set will consist of the component parts specified in table IX. The Government patterns shall be followed except for lasting allowance, which shall be determined by the contractor, and, as an alternative, the inside and outside quarter may be slightly modified to remove the vamping spring in the pattern. The insole patterns may be altered at the heel end to the extent necessary to remove excessive stock. The long side of the leather lining may be used with an inverted "V" removed in the shoe back center. The height of the cut for this inverted "V" shall be a maximum of 1-1/4 inches from the bottom of the lasting allowance on a size 4 and grade upward to a maximum height of 1-1/2 inches on size 15. The contractor shall furnish wood insole rounding patterns of dies conforming to the paper patterns furnished by the Government.

TABLE IX. Patterns and dies by width

Component parts	Contractor's dies and patterns
Patterns:	
Vamp	Whole and half sizes
Quarter inside	Whole and half sizes
Quarter outside	Whole and half sizes
Tongue 1/	Eight sizes
Vamp lining 1/	Eight sizes
Quarter lining inside	Whole and half sizes
Quarter lining outside	Whole and half sizes
Tongue lining 1/	Eight sizes
Vamp doubler 1/ 2/	Eight sizes
Quarter doubler inside 1/ 2/	Eight sizes
Quarter doubler outside 1/ 2/	Eight sizes
Box toe	Eleven sizes
Shank board	Seven sizes, widths not applicable
Counter	Eight sizes
Insole rounding	Whole and half sizes
Markers:	
Vamp 1/	Eight sizes

1/ XN has seven sizes only.

2/ Shall be used only when type I calfskin leather is used.

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3.6 Lasts. The shoes shall be made on standard U.S. MIL-1(MIL-L-3931) men's dress oxford lasts, all necessary sizes of which will be loaned to contractors by the Government.

3.7 Construction.

3.7.1 Cutting uppers. The uppers shall be cut from leather as specified in 3.3.1.1 and 3.3.1.2 and conform to the thickness specified in table X. The cut uppers shall be assembled in pairs of color, weight, and compatibility of parts. Vamps should be cut within the bend area. Belly vamps will not be accepted. No stretchy vamps or parts cut off stretch shall be permitted.

TABLE X. Thickness, uppers

	Kip and cattlehide		Calfskin	
	Minimum ounces	Maximum ounces	Minimum ounces	Maximum ounces
Vamp	3-1/2	4-1/2	2	4
Quarter	3-1/4	4-1/2	2	4
Tongue	3-1/4	4-1/2	2	4
Quarter lining	2-1/2	3-1/2	2-1/2	3-1/2

3.7.2 Skiving. The skiving of upper leather parts shall conform to the requirements specified in table XI.

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TABLE XI. Skiving requirements

Part	Side	Location	Leather scarf width (inches)							Scarf edge thickness ± 0.1 mm
			4-1/2 oz leather	3-3/4 oz leather	3-1/2 oz leather	3-1/4 oz leather	2-3/4 oz leather	2-1/2 oz leather	2 oz leather	
Quarters	Flesh	Top and front 1/	3/8 ± 1/16	*	*	13/64 ± 1/32	*	*	5/32 ± 1/32	0.7
Quarters	Flesh	Vamp line	3/16 ± 1/32	*	*	3/32 ± 1/64	*	*	5/64 ± 1/64	0.8
Vamps	Flesh	Throat and wings	1/4 ± 1/16	13/64 ± 1/32	*	*	3/16 ± 1/32	*	*	0.3
Tongues	Flesh	Bottom edge	1/4 ± 1/16	*	*	11/64 ± 3/64	*	*	5/32 ± 1/64	0.3
Quarter lining	Flesh	Long and short piece top, and vamp line	3/8 ± 1/8	*	*	*	*	*	*	0.75
Quarter lining	Flesh	Long piece closing edge	3/8 ± 1/8	*	*	*	*	3/64 ± 1/64	*	0.8

1/ When a uniform gage skiving machine is used, the scarf width for quarter tops and fronts shall be 3/8 inch ± 1/16 inch for all thicknesses.

NOTE: An asterisk indicates that the scarf width is the same as indicated in the preceding column.

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3.7.3 Marking quarter lining. The contractor's name, number, and date of contract, purchasing office, and correct size and width of the shoe shall be legibly stamped with marking ink on the outside quarter lining of each shoe.

3.7.4 Upper fitting. The upper fitting stitch types, thread sizes, and stitches per inch shall conform to the requirements specified in table XII. The stitch types specified in table XII shall conform to FED-STD-751 (see 3.7.36). Back seams shall be closed with one row of stitching closely positioned to the edge of the quarters, and rubbed down. The dog-ear of the outside quarter shall be lapped over the top of the inside quarter and stitched with one row. The top and front edge of the quarters shall be cemented with any adhesive specified in 3.3.3 and folded to enable being caught by stitching. The quarter top tape or monocord nylon thread specified in 3.3.4.4 shall be inserted in the top folding operation.

TABLE XII. Upper stitching requirements

	Stitch type	Number of stitches per inch	Threaded size	
			Needle	Bobbin or looper
Quarter lining closing	301 or	14-16	B	B
	401	14-16	B	AA
Quarter lining "V" notch (when used)	301 or	8-10	B	B
	401	8-10	B	AA
Top stitching	301	14-16	B	B
Tongue lining stitching	301	14-16	B	B
Tongue attaching	301	14-16	E	E
Back seam closing	401	11 minimum	E	B
Vamping	301	14-16	B	B
Barring	301	14-16	B	B

3.7.5 Reinforcing doubling. When reinforcing doublers are used (see 3.3.4.3), the correct size doublers shall be smoothly and securely cemented to all vamps and quarters at all points with any adhesive specified in 3.3.3 which will not harden the leather. The quarter doublers shall be positioned closely to the front edge of the quarters to enable reinforcement of the eyelets.

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3.7.6 Quarter lining and top stitching. The quarter linings shall be lapped and stitched with two rows of stitching spaced 1/8 inch ± 1/32 inch apart. If inverted "V" seam lining is used, the "V" seam shall be closed upwards from the lasting edge. The closing shall be done with one row of stitching. This seam shall extend 1/16 inch to 1/8 inch on the flesh side and the seam shall be rubbed down. The eyelet stays, when used (see 3.3.4.7), shall be securely adhered to the quarter linings or quarters. The assembled linings shall be top stitched to the quarters with a single row of stitching using a round point needle which shall pass through the quarter lining. Full advantage of the curves on the top stitching machine must be taken since the pattern is form fitting and will not allow use of flat bed under trimming machine. The quarter lining shall be sufficiently undertrimmed to minimize exposure of the lining.

3.7.7 Tongue fitting. The tongue linings specified in 3.3.4.5 shall be cemented with any adhesive specified in 3.3.3 and placed with an equal spacing around the edge of the tongue and uniformly stitched with one row of stitching on the sides and top. When used, the tongue linings cut from single napped cotton flannel specified in 3.3.4.3 shall be placed with the napped side of the material against the flesh side of the leather tongue and stitched as indicated above. The tongue shall then be placed between the leather vamp (or doubler when used) and vamp lining and stitched completely across with two rows of stitching spaced 1/8 inch ± 1/32 inch apart at center of vamp throat.

3.7.8 Eyeletting. On each quarter, five invisible eyelets specified in 3.3.14 shall be securely and smoothly clinched. The edge of the eyelets shall be 5/16 inch, + 1/16 inch, - 1/8 inch from the edge of the quarter and spaced evenly from the blucher nose to the top of the quarter.

3.7.9 Lacing for lasting. The machine lacing for lasting shall provide a 9/16 inch + 1/16 inch opening after lasting. A minimum of three pairs of eyelets shall be laced.

3.7.10 Vamping, barring. The vamping shall be done with a single row of stitching close to the edge of the quarter and a single row 1/4 inch, + 1/16 inch - 1/32 inch, from the first row. Two rows of barring, 3/8 inch by 1/8 inch ± 1/16 inch shall be included inside the vamping rows. As an alternate, the barring may be accomplished by the automatic method with tying or locking ends and the barring shall not exceed 1/2 inch in length and 1/8 inch in width. Barring may be accomplished not more than 1/8 inch above the inside vamping row to provide for one continuous vamping and barring on single needle machine. When modified quarter patterns are used, single needle computerized or numerically controlled stitching may be used.

3.7.11 Sole leather stock fitting. The outsoles shall be sorted to meet document requirements. Any adhesive specified in 3.3.3 shall be applied to the flesh side of the outsole.

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3.7.12 Insole preparation. The leather insoles shall be sorted for varying fibers to obtain uniform channeling and cased for even weight. The insoles shall be fleshed and rounded to patterns as specified in 3.5. Die cut insoles will be accepted provided they conform to last bottom paper patterns loaned by the Government. A ± 1/64 inch tolerance from Government loaned insole patterns will be allowed.

3.7.13 Insole scoring. When leather insoles are used, the lip scoring cut shall not penetrate to a depth exceeding 50 percent of the insole thickness and shall be located 2-9/16 inches ± 1/16 inch from the heel end of a size 8 insole and shall grade from that size at the rate of grade obtained from a lip cutting and scoring machine. When Texorist insoles are used, in lieu of scoring cuts, ink marking shall be used. Location measurements and grading shall be as specified above.

3.7.14 Insole channeling. The insoles except stuck-on rib insoles shall be channeled and may be chamfered. The channel leaf and outside lip shall be not less than 1/3 the thickness of the insole, and the channel leaf shall be slightly thicker than the lip. The in-between substance shall be 1/64 inch to 1/32 inch greater than the combined thickness of the leaf and lip. The channel margin shall be a minimum of 5/32 inch, and the maximum shall not exceed the required extension on the finished shoe (see 3.7.29). The lips shall be cemented with a latex adhesive specified in 3.3.3 (a) or (b) and pressed together at lip setting to incline inward and be a minimum of 1/8 inch in height.

3.7.14.1 Assembly of stuck-on rib insoles. When assembled, the stuck-on rib insoles shall meet the requirements of 3.3.4.6.

3.7.15 Insole backing. All insoles except stuck-on rib insoles (see 3.3.4.6.1) shall be reinforced by the application of cotton duck specified in 3.3.4.1. The duck shall be coated on one side with any adhesive specified in 3.3.3 and bonded to the flesh side of the insole. The duck shall cover the entire area between the ribs from the score line forward, shall extend to the top of the upstanding rib, shall be firmly and smoothly fitted and bonded to the base and the lip, and shall be trimmed to the required rib height. The insoles shall be cemented with any adhesive specified in 3.3.3 on the flesh side prior to attaching the coated duck. The 3/4-inch tape reinforcing method may be used as an alternative using a cotton duck tape conforming to the requirements of 3.3.4.1.

3.7.16 Lasting. The lasts, insoles, counters, and box toes shall be assembled in accordance with the size and width of the uppers. Uppers may be conditioned by any suitable means except they shall not be dipped in water. Insoles of the correct size and width shall be tacked to the last with not less than five tacks: one in the center of the heel seat, one at the shank, one at each side of the ball, and one at the toe. In lieu of tacks, staples may be used for attaching the insole to the last. Edges of the insole shall be flush with the last bottom at all points. The heel seat of the insole shall be smooth and even with the heel seat of the last. The shoe shall be

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assembled to the last using tacks only in the heel seat. A light coating of any adhesive specified in 3.3.3 shall be applied between the vamp doubler, when applicable, and the vamp lining at the inside shank of each shoe. The counters shall be assembled in accordance with the schedule specified in 3.3.2.1 and cemented on both sides. When leather counters are used, the flesh side of the leather counter shall be toward the inside of the shoe. The correct size proxylin or polystyrene type box toe specified in 3.3.6 shall be conditioned and inserted between the leather vamp or vamp doubler, when applicable, and the vamp lining. The uppers shall be assembled to the lasts in a manner to provide for a wiped-in heel seat of approximately 1/2 inch, and drawn over the lasts with a strong tension on the pulling-over machine to assure that the quarters at the blucher points and the vamps are down over the last and the blucher noses are even. The sides of the shoes shall be spindled, drawing the upper snugly to the last and then side-lasted so that, when stapled, the upper will be held securely to the last. The vamp lining shall be pulled smooth and tight without any tears. The heel seat shall be wiped-in flat and free from wrinkles. The toe shall be smoothly wiped-in and securely attached around the base of the insole rib. As an alternative, any suitable equipment or method may be used to smoothly and securely attach the toe around the base of the insole rib.

3.7.17 Time allowance on lasts. The shoes shall remain on the lasts until thoroughly dry.

3.7.18 Inseaming. Inseaming shall be done by machine with the thread specified in 3.3.5.3 thoroughly waxed, with tension on the thread, using a needle not larger than No. 41 and with not less than 3-1/4 stitches to the inch. The welting specified in 3.3.1.5 shall be tempered to provide a tight seam without cracking the welt. The welting shall be inseam stitched to the bottom of insole rib, from insole rib end to rib end, and shall catch in counter with one or more stitches on each end. As an alternative, the operations for side-lasting and inseaming may be performed simultaneously with any suitable equipment or method. Prior to heel seat lasting and inseaming, each shoe shall be spindled, have the forward edge of the counter pulled into position, and have the upper, with its lining, drafted snugly down to the last on both sides of the shoe.

3.7.19 Tack and staple pulling, inseam trimming. All tacks and staples shall be removed from the bottom of insole and no broken tack or staple points shall remain. The inseam shall be closely trimmed from butt to butt of welt without cutting or damaging the stitches. The ends of the welt shall be skived with a 5/8 inch $\pm$ 1/8 inch bevel tacked within the butt area. The welt shall be beaten out smoothly while in temper to lie flat around the entire forepart of the shoe.

3.7.20 Shank fitting, bottom filling. The shanks shall be selected for applicable size in accordance with the shank fitting schedule specified in 3.3.8.3. The shanks shall be inserted in position with the shank board filling the cavity between the edges of the vamp and the quarter from the ball line rearward to the back of the heel seat. The forward end of the shank

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shall be flush with the insole and shall fit the contour of the shoe bottom back of the ball line. The shank shall be attached to the shoe with pitch, wax, or two tacks or staples, one at each side of the shank board piece at the rear end of the shank board. The bottom filler shall be firmly pressed into the insole channel from the toe, extending to the forward end of the shank with a uniform and even surface. The bottom shall present a flat, smooth surface for sole laying. Any excess cavities at the heel portion shall be filled with bottom filler.

3.7.21 Sole laying. The shoe bottoms shall be cemented from toe end to heel breast line with any adhesive specified in 3.3.3, and the outsoles shall be laid on a sole laying machine with pressure to hold the sole to the welt at all points. The soles shall be laid evenly and shall be large enough to obtain the edge extension specified in 3.7.29.

3.7.22 Rough rounding. The soles shall be uniformly rounded on a rough rounding machine to provide for the edge extension of sole and welt specified (see 3.7.29).

3.7.23 Sole stitching. The outsoles and welts shall be stitched together on a lockstitch machine, using waxed polyester thread specified in 3.3.5.2. A needle and awl not larger than No. 47 shall be used. The stitching of the leather outsole shall be in a groove 7 to 9 stitches to the inch, with the stitches laid on the surface of the welt, close to the outer edge of the welt, and the stitches shall extend under the heel on the finished shoes. The width and depth of the groove shall be such that the top surface of the thread is flush with or slightly lower than the surface of the leather outsole. The lock shall be placed just under the surface of the outsole (see 3.7.36.2). When rubber outsoles are used, the groove is not required and there shall be 6 to 7 stitches per inch. The stitches shall be laid on the surface of the welt, close to the outer edge of the welt on the finished shoes. The lock shall be placed just under the surface of the outsole.

3.7.24 Heel seat fastening. The heel seat fastening shall be done by a fiber fastening machine using fiber pegs specified in 3.3.13. When fastening is accomplished using automatic (gang fastener) equipment, the pattern provided by the equipment will be acceptable. When the pegs are driven one at a time, there shall be not less than four to the inch. When used, steel or brass nails shall be spaced not less than three to the inch and shall clinch properly on the insole. Pegs and nails shall be positioned at the edge of the insole and extend around the heel seat from welt butt to welt butt.

3.7.25 Heel seat rounding. The heel seat of the outsoles shall be smoothly rounded from welt butt to welt butt.

3.7.26 Leveling. Bottoms shall be progressively leveled while in temper on a leveling machine by means of metal rolls. The machine shall be adjusted to level bottom of the sole from toe end, back to breast line. As an

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alternative, any suitable equipment may be used that will adequately level the sole bottom while in temper. When rubber outsoles are used, the leveling of bottoms is not required.

3.7.27 Heeling. The heel and base shall be attached with eight nails, or with seven nails with the center breast nail omitted. The nails shall be long enough to ensure a secure clinch on the insole. The heeling machine shall be equipped with proper length drivers to drive the heel nails evenly against the metal washers in the heels and clinch the heel nails on the insole.

3.7.28 Heel trimming and finishing. The edges of the sole and heel shall be shaved square with a close bevel on the top edge of the outsole around the heel area and then scoured to a smooth finish free of indentations and ridges. The heel breast shall be scoured. The heel seat and base shall be stained to match the upper and finished with wax to a bright finish.

3.7.29 Edge trimming. The edges shall be trimmed square by a cutter having a flat bed with no beads. The finished extension of outside ball shall be not less than 3/16 inch, and the finished extension of toe and inside ball not less than 1/8 inch when measured at right angles to the upper. The jointed edges of the sole and welt shall be smooth.

3.7.30 Edge setting. A filler shall be applied to the edge, allowed to dry, then the edge shall be stained black with the ink specified in 3.3.17 and the ink allowed to dry. The edges shall then be set to a smooth bright finish on an edge setting machine using heat (not frictionized) edge irons. When rubber outsoles are used, the edges of the welt and soles shall be finished with ink specified in 3.3.17. The ink may be self-setting or may be set on an edge setting machine to a hard bright finish.

3.7.31 Bottom finishing. The bottoms shall be clean and smooth.

3.7.32 Finish.

3.7.32.1 Preparation. The shoes shall be cleaned, removing the accumulated dirt, wax, cement, and other foreign matter. All thread ends shall be trimmed.

3.7.32.2 Treeing. The shoes shall be treed while on the U.S. MIL-1 oxford lasts, or on a treeing machine using right and left tree feet conforming to the U.S. MIL-1 oxford last. All wrinkles shall be removed and no material shall be used that may injure the leather or thread.

3.7.32.3 Final finish. The shoes shall be repaired, filled, and given a bright finish by the sponge or spray method with materials and methods specified in 3.3.18 through 3.3.20 to achieve a bright luster. The quarter vamp lining raw edges shall be stained to match the upper leather.

3.7.32.4 Edge pad and brush. The edges of the sole and heel shall be padded and brushed to a bright finish.

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3.7.33 Nails, tacks, and staples. Nails, tacks, and staples that have been left protruding through the insole and cannot be pulled out, shall be cut close to the surface leaving no protruding stumps. A mechanical tack detector or other suitable method may be used to indicate the presence of any protruding tacks or staples inside the shoe.

3.7.34 Heel pads. Pads shall be inserted grain side up. The heel pads shall be coated, with any adhesive specified in 3.3.3, over the surface of the flesh side and shall be pressed firmly and completely into the heel seat. Heel pads shall be fitted in accordance with size schedule.

3.7.35 Lacing-mating. The shoes shall be mated and a lace shall be inserted through the lower eyelet of each shoe and tied in a loose bow knot.

3.7.36 Repairs of stitching.

3.7.36.1 Type 301 and 401 upper fitting stitches. Repairs of type 301 and 401 stitches shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching one stitch in back of the end of stitching.

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the shoe (in-process or end item) shall be repaired by overstitching. The stitching shall start one stitch in back of the defective area, continue over the defective area, and continue one stitch beyond the defective area onto the existing stitching.

3.7.36.2 Inseaming and sole stitching. Repairs of inseaming and outsole stitches shall be in accordance with 3.7.36.1 and in addition, thread cuts occurring as a result of trimming operations shall be repaired in the same manner as specified for thread breaks.

3.7.37 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2.2, 4.4.2.3, and 4.4.4 shall be removed from production and replaced with non-defective and properly matched components.

3.8 Workmanship. The finished shoes shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

* 4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table XVI. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition, testing shall be performed on components and materials listed in table XIII for the characteristics shown. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table XIII, they shall be as required by the referenced test method. The lining leather sampling procedure shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more sample units, the lot average, or the composite sample fail to meet any requirement specified. The sample size (except for lining leather) shall be as follows:

Lot size	Sample size
800 or less	2
801 to and including 22,000	3
22,001 and over	5

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TABLE XIII. Component tests

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	No. of determinations per sample unit	Results reported as	Sample unit
Lining leather (calf, kip, bellies, or side leather)	Paranitrophenol content	3.3.1.2	6711 3/	-	1/	-	-	4.4.1
	Tannage	3.3.1.2	2/	-	-	-	-	
	Thickness	3.3.1.2	1011 3/	X	-	-	-	
Heel pad (one pad)	Paranitrophenol content	3.3.1.3	6711 3/	-	1/	-	-	Five heel pads
	Material identification	3.3.1.3	2/	-	-	-	-	
	Thickness	3.3.1.3	1011 3/	X	-	-	-	
	Tannage	3.3.1.3	2/	-	-	-	-	-
Insole (Texorist)	Copper-8-quinolinolate	3.3.1.4.2	2/	-	-	-	-	-
Adhesive	Material identification	3.3.3	2/	-	-	-	-	-
Cotton vamp and quarter doubler (for type I calfskin leather only) (1 yard full width)	Material identification	3.3.4.3	2/	-	-	-	-	1-yard full width
	Weight	3.3.4.3	5041 4/	X	-	-	-	
	Yarns per inch	3.3.4.3	5050 4/	X	-	-	-	
	Breaking strength	3.3.4.3	5100 4/	X	-	-	-	
	Non-fibrous material	3.3.4.3	2611 4/	X	-	-	-	
Tape, quarter top (1 gross yard)	Material identification	3.3.4.4	2/	-	-	-	-	1 yard
	Breaking strength	3.3.4.4	5100 4/	X	-	-	-	

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TABLE XIII. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	No. of determinations per sample unit	Results reported as	Sample unit
Wool felt tongue lining (1 yard full width)	Weight	3.3.4.5	5041 4/	X	-	5	Nearest ounce	1-yard full width
	Wool content	3.3.4.5	2101 4/	X	-	-	-	
	Water soluble content	3.3.4.5	2611 4/	X	-	1	Nearest 0.5 percent	
Insole ribbing fabric (1 yard)	Material identification	3.3.4.6.1	2/	-	-	-	-	-
	Weight	3.3.4.6.1	5041 2/ 4/	-	-	-	-	-
	Yarns per inch	3.3.4.6.1	5050 2/ 4/	-	-	-	-	-
	Breaking strength	3.3.4.6.1	5100 2/ 4/	-	-	-	-	-
Box toe	Material identification	3.3.6	2/	-	-	-	-	-
	Impregnation compound identification	3.3.6	2/	-	-	-	-	-
	Thickness	3.3.6	2/	-	-	-	-	-
Bottom filler composite (1 pound)	Material identification	3.3.7.1 and 3.3.7.2	2/	-	-	-	-	-
	Proportion of cork to binder	3.3.7.1 and 3.3.7.2	2/	-	-	-	-	-

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Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	No. of determinations per sample unit	Results reported as	Sample unit
Bottom filler binder only (1 pint)	Softening point of binder	3.3.7.1 and 3.3.7.2	E 28 5/	-	X	2	Avg of 2 determinations to nearest 1°F	1-pound composite
	Penetration of binder	3.3.7.1	D 5 5/	-	X	3	Avg of 3 determinations to nearest mm	
	Solubility of binder in water	3.3.7.1 and 3.3.7.2	2/	-	-	-	-	
Shank board	Material identification	3.3.8.1.1	2/	-	-	-	-	-
	Thickness	3.3.8.1.1	2/	-	-	-	-	-
Steel shank (1 pair)	Material identification	3.3.8.1.2	2/	-	-	-	-	One steel shank
	Hardness (Rockwell C scale)	3.3.8.1.2	E 18 5/	X	-	-	Nearest number	
	Thickness and width	3.3.8.1.2	2/	-	-	-	-	
	Coating identification	3.3.8.1.2	2/	-	-	-	-	
Fiberglass shank	Material identification	3.3.8.2	2/	-	-	-	-	-
	Width	3.3.8.2	2/	-	-	-	-	-
	Heat curable	3.3.8.2	2/	-	-	-	-	-
Heel base rubber (1 pair)	Material identification	3.3.10.1	2/	-	-	-	-	One pair
	Hardness	3.3.10.1	6/	-	X	3	Nearest	

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Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	No. of determinations per sample unit	Results reported as	Sample unit
Heel base plastic (1 pair)	Material identification	3.3.10.2	2/	-	-	-	-	One pair
	Specific gravity	3.3.10.2	D 792 5/	X	-	2	Pass or fail	
	Hardness	3.3.10.2	7/	-	X	3	Nearest number	
	Unfilled	3.3.10.2	2/	-	-	-	-	
Peg, heel seat	Material identification	3.3.13	2/	-	-	-	-	-
	Diameter	3.3.13	2/	-	-	-	-	-
Eyelet (1 pound)	Material identification	3.3.14	2/	-	-	-	-	Two eyelets
	Finish	3.3.14	2/	-	-	-	-	
	Thickness	3.3.14	2/	-	-	-	-	
	Outside diameter of flange	3.3.14	Gage	X	-	2	Nearest .001 inch	
	Overall length before setting	3.3.14	Gage	X	-	2	Nearest .001 inch	
	Diameter of hole before setting	3.3.14	Gage	X	-	2	Nearest .001 inch	
Wax	Material identification	3.3.16	2/	-	-	-	-	-

1/ Requirement applicable to "composite."

2/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

3/ Refers to FED-STD-311.

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4/ Refers to FED-STD-191.

5/ Refers to ASTM Standard.

6/ Shore Durometer or Rex Gauge to be used.

7/ Shore Durometer Gauge D Scale to be used.

4.4.1.1 Certification. The shoe manufacturer shall furnish a certificate of compliance for the top spray requirements specified in 3.3.1.1 and identify the top finish used.

4.4.1.2 Examination of Texorist insoles (when used). The Texorist insoles shall be examined for the defects listed below. The lot size shall be expressed in units of one insole. The sample unit shall be one insole. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

Any hole, cut, tear, or gouge.
Any brittle area or evidence of delamination.
Thickness not as specified.
Sole outline does not conform to required pattern.
Not cut in specified direction.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 In-process examination of vamps prior to fitting. Every cut vamp shall be examined for the defects listed below prior to fitting. Any cut vamp containing one or more defects shall be rejected.

Scratch or scar
Slaughter cut
Flesh cut penetrating through the grain surface
Damaged grain surface
Brand
Belly cuts
Flanky, pipey, loose, cracked, or any rough or coarse grain
Cluster of veins, split veins, or pronounced veins

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4.4.2.2 In-process examination of uppers after all fitting. The upper assembly shall be examined for the defects listed in table XIV. The lot size shall be expressed in units of one upper. The sample unit shall be one completely fabricated upper assembly prepared for lasting. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

NOTE: Defects designated by an asterisk (*) in table XIV shall be scored as major when seriously affecting serviceability and minor when affecting serviceability but not seriously.

TABLE XIV. Defects of uppers after all fitting

Examine	Defect	Classification Major	Minor
Construction and workmanship	Any component missing or not specified type	X	
	Vamp or quarter doubler (when applicable) not securely cemented to leather		X
	Any component misplaced, damaged, or not affixed as specified	*	
	Construction not as specified	*	
	Backseam not rubbed down		X
	Quarter top tape or monocord nylon thread not inserted in fold at the top of quarter		X
	Vamp not crimped		X
Quality of leather	Thickness more than 1/2 ounce less than the minimum specified	X	
	Thickness more than the maximum or up to 1/2 ounce less than minimum specified		X
	Slaughter cut	*	
	Off stretch cut	X	
Stitching of upper quarter lining V notch	Quarter lining V notch has less than 8 or more than 10 stitches per inch		X

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4.4.2.3 In-process examination of shoe before bottom filling. The partially fabricated shoe shall be examined for the defects listed in table XV. The lot size shall be expressed in units of one shoe. The sample unit shall be one partially constructed shoe at a point after lasting and attachment of shank but before the application of the bottom filler. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

NOTE: Defects designated by an asterisk (*) in table XV shall be scored as major when seriously affecting serviceability and minor when affecting serviceability but not seriously.

TABLE XV. Defects examined before bottom filling

Examine	Defect	Classification Major		Minor
Bottom of shoes	Any component missing or not specified type	X		
	Shank not properly positioned or not fitting contour of shoe bottom		*	
	Shank wrong size or malformed			X
	Shank not securely attached			X
	Any tear in cotton duck in stitch area:			
	- more than 3/4 inch	X		
	- more than 1/2 inch but not more than 3/4 inch			X
	Insole tack, staple, or anchor tack not removed	X		
	Excess leather in heel seat interfering with proper fit of shank	X		
	Upper damaged in lasting		*	
	Poor heel seat, side, or toe lasting		*	
	Inseam not properly trimmed		*	
	Less than three stitches per inch on inseam	X		
	Less than 3-1/4 inch but not less than 3 stitches per inch on inseam			X
	Inseam stitches broken, two or more consecutive skipped inseam stitches, or inseam stitches cut or damaged during trimming operation:			
	- not repaired	X		
	- not repaired as specified			X
	Broken insole rib		*	

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TABLE XV. Defects examined before bottom filling (cont'd)

Examine	Defect	Classification Major	Minor
Bottom of shoes (cont'd)	Inseam stitching not at bottom of insole rib, i.e., 1/16 inch from bottom of rib for 6 or more stitches		X
	Inseam stitching not at bottom of insole rib, i.e., 1/8 inch from bottom of rib for more than 4 stitches	X	
	Inseam stitching not in welt groove	X	
	Ends of counter not caught in inseam stitching		X
	Welt not properly beaten out, i.e., not flat		X
	Welt butts not skived	X	
	Welt butts not tacked within butted area		X
Upper part of shoes	Uppers not tightly pulled down to last		X
	Lace opening less 1/2 inch or more than 5/8 inch	X	
	Quarter not laced in three pairs of eyelets		X

4.4.2.4 In-process testing (stuck-on rib insole). When used, the stuck-on rib shall be tested for compliance with the requirements of 3.3.4.6.2. Method 2061 of FED-STD-311 shall be used for shear strength test. Method 2171 of FED-STD-311 shall be used for stitch tear strength. The sample unit shall be two ribbed insoles, one for each test, and the sample size shall be five, regardless of the lot quantity. Requirements are applicable to the sample unit for minimum unit requirements and lot average for average requirements. The lot shall be unacceptable if one or more sample units or the lot average of all units fail to meet the requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

4.4.3 End item critical defect examination. Every end item, prior to the sampling examination in 4.4.4, shall be examined for critical defects. Any protruding point of a lasting staple, nail, or tack forward of the heel breast line shall be a critical defect. Any end item found to contain one or more critical defects shall be rejected.

4.4.4 End item visual examination. The shoes shall be examined for the defects listed in table XVI. The lot size shall be expressed in units of one shoe. The sample unit shall be one completely fabricated shoe and the selection

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shall be by pairs. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects. Defects of pairing shall be classified as a single defect.

NOTE: Regarding conditions affecting appearance, defects shall be scored only when the condition is plainly visible at a distance of 3 feet or more. Defects designated by an asterisk (*) in table XVI shall be scored as major when seriously affecting serviceability or appearance and minor when affecting serviceability or appearance but not seriously.

TABLE XVI. End item visual defects

Examine	Defect	Classification Major	Minor
Pairing	Not properly mated, i.e., not right and left of same size	X	
	Variation in color or appearance	*	
	Box toe crooked, long, or short	*	
Cleanness	Any non-removable spot, stain, or foreign matter affecting appearance		X
Color and finish	Not color specified	X	
	Color not uniform affecting appearance		X
	Upper leather not finished to specified brightness		X
	Sole and heel edges not padded and brushed to a bright finish		X
	Finish runs or is streaky, chipped, or flaky on upper		X
	Quarter vamp lining raw edges not stained to match upper leather		X
Design, type, and size	Not as specified	X	
Leather, upper (quality)	Any of the defects in leather, referenced in 3.3.1.1, specified thereas to location in vamps, forward part of quarter, or in the counter and tongue areas, and any other defect in the vamps of forward part of the quarter that is visible at a distance of 3 feet or more, shall be classified as follows:		

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TABLE XVI. End item visual defects (cont'd)

Examine	Defect	Classification Major		Minor
Leather, upper (quality) (cont'd)	- when seriously affecting appearance	X		
	- when affecting appearance but not seriously			X
	Stretchy vamp	X		
Calf (type I)	Not full grain or burnished full grain smooth finish	X		
Kip (type II and cattlehide type III)	Not burnished full grain smooth finish	X		
Lining leather	Grain not smooth or uniform, seriously affecting serviceability	X		
	Slaugher cuts, seriously affecting serviceability	X		
	Pipey, flanky, or off stretch cut	X		
Construction and workmanship	Any cut, tear, hole, or abrasion		*	
	Any component or assembly omitted or misplaced, operation omitted or not properly performed (unless otherwise classified herein)		*	
	Loose leather fibers exposed along cut edge of quarter vamp line			X
Seams and stitching of upper	Open seam not repaired	X		
	Repair of open seam not as specified			X

NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur. On multiple stitched seams, a seam is considered open when either one or both sides of the seam are open. When any of the above conditions occurs on the inside of the shoe, it shall be scored as an asterisk (*) defect.

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TABLE XVI. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams and stitching of upper (cont'd)	Tight tension resulting in puckering or cutting of leather	*	
	Loose tension resulting in a loosely secured seam	*	
	Wrong seam or stitch type	*	
	Back seam has less than 11 but not less than 9 stitches per inch		X
	Back seam has less than 9 stitches per inch	X	
	Any row of stitching (except back seam) with less than 14 but not less than 12 stitches per inch		X
	Any row of stitching (except back seam) with less than 12 stitches per inch	X	
	More than the specified maximum number of stitches	*	
	NOTE: A plus tolerance of 3 stitches per inch will be allowed when stitching over heavy places or turning sharp corners.		
	Gage or spacing of stitching not as specified or irregular	*	
	Top stitching not passing through quarter lining	X	
	Stitching omitted where required	*	
	Thread ends not trimmed throughout shoe		X
	Needle holes or needle chew	*	
	Stitching overrun resulting in stitching where not intended		X
Outsole	Less than 6 stitches per inch	X	
	Less than 7 but not less than 6 stitches per inch, or more than 9 stitches per inch		X
	Less than 5 stitches per inch (applicable to rubber outsole)	X	

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TABLE XVI. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Outsole (cont'd)	Less than 6 but not less than 5 stitches per inch, or more than 7 stitches per inch (applicable to rubber outsole)		X
	Outsole stitching irregular		X
	Broken stitch or two or more consecutive skipped stitches:		
	- not repaired	X	
	- not repaired as specified		X
	Stitches short at heel breast	*	
	Stitches not in groove (applicable to leather outsole)	*	
Counter	Rolled or curled counter	*	
	Soft counter		X
Edge making	Edge trimmed into sole stitching	*	
	Edge not trimmed square or trimming is irregular affecting appearance		X
	Sole extension less than specified minimum	*	
	Edge not set	X	
	Edges not properly set, e.g., chatter marks		X
	Jointing of sole and heel trimming not smooth, e.g., ridges, undercuts	*	
Heel finishing and attaching	Heel not properly trimmed, i.e., is pronouncedly flared or tapered	*	
	Heel not finished smooth, including heel breast	*	
	Open heel seat	*	
	Checked heel, i.e., separation between heel and heel base or between heel base and outer sole		X
	Crooked or wrong size heel		X
	Cracked heel base	X	
	Any heel nail other than center breast, corner breast, or center back nail missing		X

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TABLE XVI. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Heel finishing and attaching (cont'd)	Center back or corner breast nail missing or not clinched on insole	X	
	More than one heel nail, other than center back or corner breast nail, not clinched		X
	NOTE: Evidence of heel nail on the insole shall indicate clinching.		
Eyelets	Number of eyelets not as specified but each row has the same number		X
	Not the same number of eyelets in each row	X	
	Eyelets not properly spaced within the row or misalignment between the rows to an extent interfering with proper lacing	*	
	Edge of any eyelet less than 1/8 inch from the edge of the quarter	X	
	Edge of any eyelet less than 3/16 inch but not less than 1/8 inch or more than 3/8 inch from the edge of the quarter		X
	Any eyelet not securely clinched		X
Inseaming	Grinning seam, i.e., thread exposed	X	
	Stained seam, i.e., needle holes visible but thread not exposed		X
	Dropped welting, skipped stitches, or cut inseam	X	
Insole	Short or long	*	
	Any protruding point of lasting staple, tack, or nail forward of heel breast line	1/	
	Any protruding point of lasting staple, tack, fastener peg, or nail in heel seat area	X	

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TABLE XVI. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Insole (cont'd)	More than two heel seat fasteners or nails runoff insole	X	
	More than three heel seat fasteners or nails not properly positioned to edge of insole		X
	No evidence of heel seat fastening		X
Lining	Quarter or vamp lining wrinkled, loose, or with excessive fullness	*	
	Quarter or vamp lining torn	X	
	Tongue lining wrinkled or torn		X
Tongue lining (alternate materials)	Not cut as specified		X
	Ravelled		X
Heel pad	Not firmly and adequately adhered to heel seat		X
Marking	Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible, or not specified size		X

1/ Any protruding point of lasting staple, tack, or nail forward of the breast line found in the sample shall be scored as a critical defect and shall be cause for rejection of the lot represented. The AQL specified in 4.4.4 does not apply to staple tack or nail forward of the heel breast line.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of unit packs per container is more or less than required Size indicated on shoes not the same as specified on unit packs or shipping containers 1/

1/ For this defect, one unit pack from each container in the sample shall be examined.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.5 Methods of inspection.

4.5.1 Cracking. After being subjected to a temperature of 150° ± 2°F for 7 days or 212° ± 2°F for 24 hours in a forced circulating air oven, condition the rubber heel base for a minimum of 16 hours at room temperature and then bend double from heel to breast with the cupping on the inside of the bend.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Both shoes of a pair shall be held with the bottoms together, heel to sole, and placed in a plastic film bag measuring approximately 10 inches by 20 inches made from closed cell cushioning material conforming to class 1 or 2, 1/4 inch cell height of PPP-C-795. Alternatively, each shoe shall be inserted into a snug-fitting, clear polyethylene film bag of 0.00225 inch ± 0.00025 inch film thickness. The open end of the bag shall extend a minimum of 5 inches beyond the length of the shoe. Each pair of shoes shall then be unit packed in a one-piece fold-up style shoe box fabricated as specified in 5.1.1.1 with dimensions as specified in table XVII. Shoes in individual bags shall be placed in the shoe box with the bottoms together heel to sole.

5.1.1.1 Shoe box. Boxes shall be fabricated from 125 pounds bursting test, C flute, corrugated fiberboard. The box shall be one piece, interlocking, hinged lid, with a tuck-in-tongue style closure in accordance with figure 2.

TABLE XVII. Shoe box (inside dimensions) inches (± 1/16 inch)

Range of shoe sizes	Length	Width	Depth
4 through 10-1/2, all widths	13	6-1/4	5
11 through 15, all widths	14-7/8	6-1/4	5-1/2

5.1.2 Commercial. Shoes shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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* 5.2.1 Level A packing. Twelve pairs of shoes of one size and width only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, class weather-resistant grade V2s of PPP-B-636. Level A unit packs shall be packed on end, four in length, three in width, and one in depth within a shipping container. Inside dimensions of shipping containers shall be approximately 21 inches in length, 20-1/2 inches in width, and 13-1/2 inches in depth for sizes 4 through 10-1/2 shoes, and 23 inches in length, 20-1/2 inches in width, and 15-1/2 inches in depth for sizes 11 through 15 shoes. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5. Voids within the shipping container shall be filled with folded or scored corrugated fiberboard pads to immobilize the contents. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Twelve pairs of shoes of one size and width only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed on end, four in length, three in width, and one in depth within the shipping container. Inside dimensions of shipping containers shall be approximately 21 inches in length, 20-1/2 inches in width, and 13-1/2 inches in depth for sizes 4 through 10-1/2 shoes and 23 inches in length, 20-1/2 inches in width and 15-1/2 inches in depth for sizes 11 through 15 shoes. Approximate dimensions are furnished as a guide only. Voids within the shipping container shall be filled with folded or scored corrugated fiberboard pads to immobilize the contents. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial. Shoes, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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* 5.3 Palletization. When specified (see 6.2), shoes packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern 6 shall be used for sizes 4 through 10-1/2 shoes, and pallet pattern 90 shall be used for sizes 11 through 15 shoes. For pallet pattern 6, interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Width marking. Following the nomenclature and size designation in the same size lettering, the width shall be spelled out; i.e., "EXTRA NARROW," "NARROW," "REGULAR," "WIDE," OR "EXTRA WIDE."

6. NOTES

6.1 Intended use. The shoes are intended for wear by male personnel of the Department of Defense.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Sizes and widths required (see 1.2).
c. When a first article is required (see 3.2, 4.3, and 6.3).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Guide sample. For access to samples, address the contracting activity issuing the invitation for bids.

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* 6.5 Top spray finishes. The following water base, wax acrylic resin finishes have been found acceptable (see 3.3.1.1):

Top spray finish	Manufacturer
WTS-1562 WPS-910	Paule Chemical Company P.O. Box 3569 Corwin Street Peabody, MA 01960

6.6 Fiberglass shank. A fiberglass shank manufactured by American Shoe Machinery Co., 199 Newbury Street, P.O. Box 390, Danvers, MA 01923 under their designation "TRU-FIT" has been found to meet the requirements of 3.3.8.2.

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.3).

* 6.8 Subject term (key word) listing.

Footwear
Men's
Shoes, dress
Shoes, oxford

6.9 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army - GL
Navy - NU
Air Force - 11

Review activities:

Army - MD
Navy - MC
Air Force - 99, 82
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

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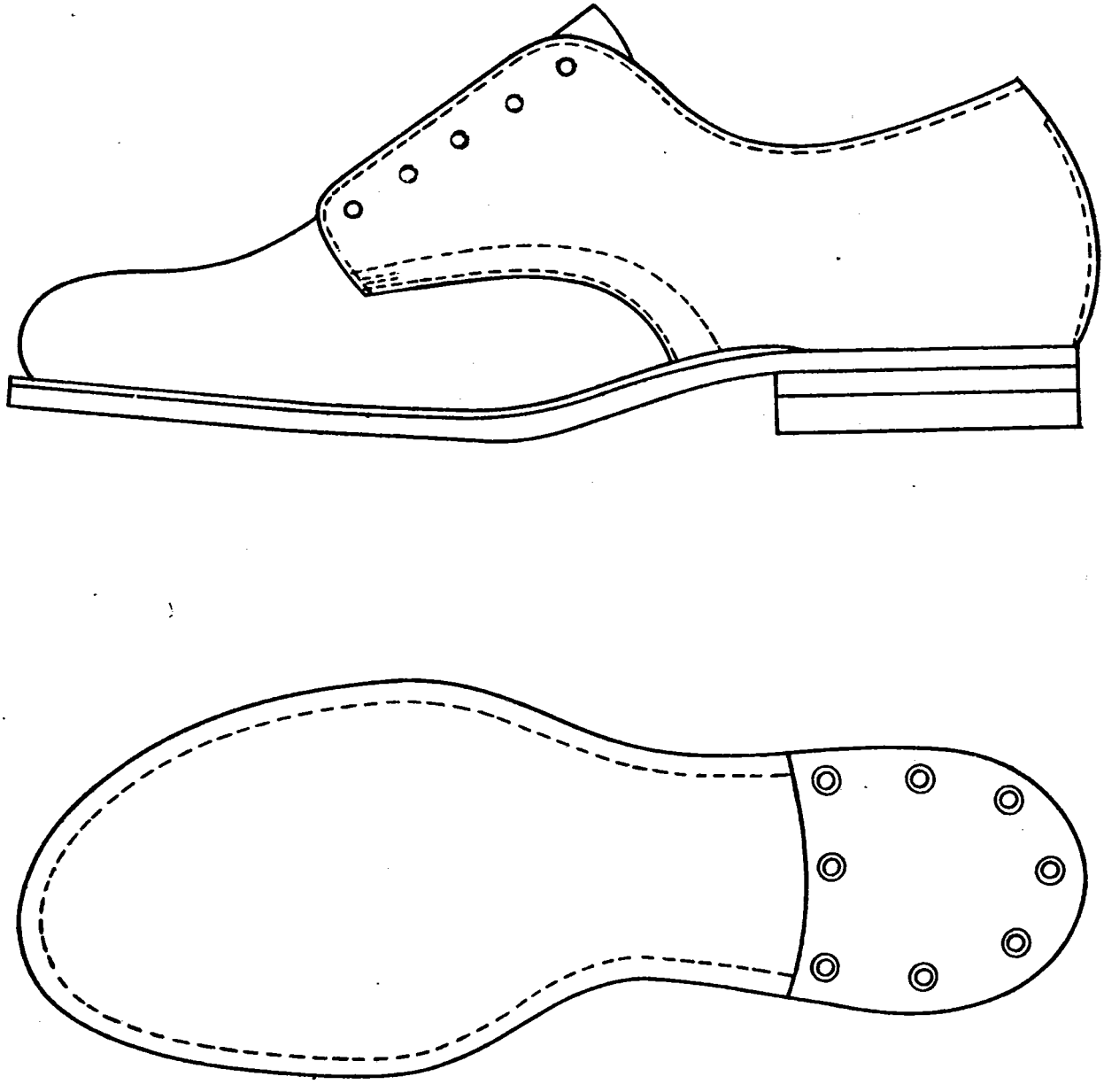


FIGURE 1. Shoe, Men's, Dress, Oxford.

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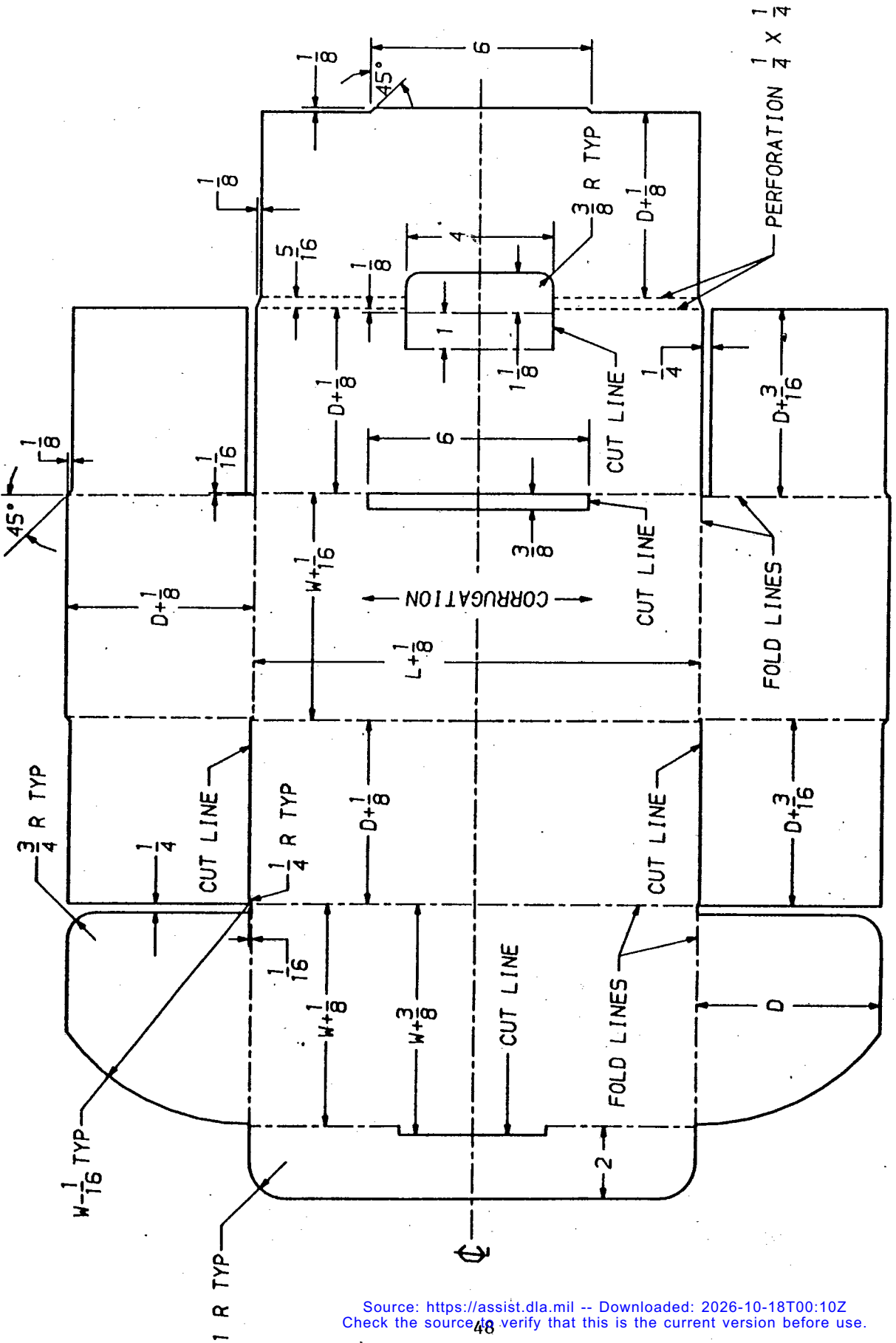


Figure 2. Box, shoe, corrugated fiberboard One Piece Fold-Up.

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